

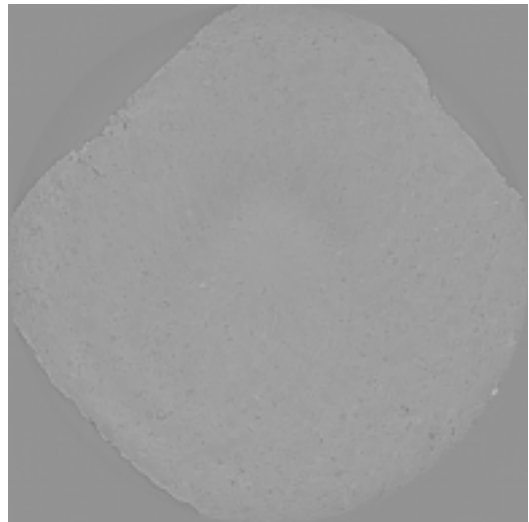
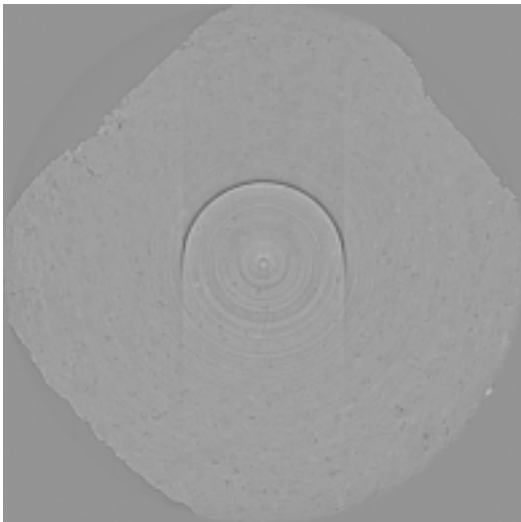


UNIVERSITY OF GENEVA

Faculty of Science

Assignment 4

Computational Imaging
14x062



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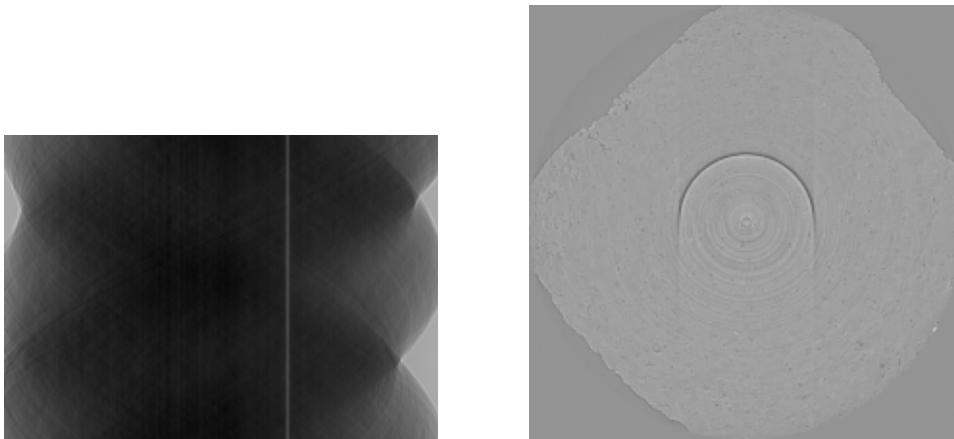
1 Introduction

The computational imaging allows to reconstruct an image from a set of measurements from different kind of devices, such as X-ray devices. The computed tomography (CT) is a specific type of computational imaging that allows to reconstruct a 3D image from a set of 2D projections, called sinograms. The reconstruction of the image from the sinogram is a challenging problem, especially when the sinogram is corrupted by noise or artifacts, such as stripe artifacts. In this report, we will explore different methods to remove stripe artifacts from sinograms and reconstruct the image.

2 Methods

In CT, a X-ray source rotates around the object and collects projections at different angles, which are then used to reconstruct the image. All the projections are collected in a sinogram, which is a 2D image where the horizontal axis represents the angle of the projection and the vertical axis represents the position of the detector.

The stripe artifacts are caused by the presence of defects in the detector such as dead pixels or hot pixels, which can be due to various reasons, such as dust, scratches, or electronic noise. These artifacts appear as vertical stripes in the sinogram as shown on Figure 1a, which can significantly degrade the quality of the reconstructed image, like in Figure 1b. Indeed, the presence of stripe artifacts can seriously affect the quality of the diagnosis due to the presence of ring artifacts in the reconstructed image, which can be misinterpreted as pathological features or hide real pathologies.



(a) Sinogram with stripe artifacts (b) Reconstruction without stripe removal

Figure 1: Original sinogram with stripe artifacts and reconstruction without removal

To remove the stripe artifacts, we will explore different methods, such as Fourier-based methods, wavelet-based methods, normalization-based methods, regularization-based methods, and sorting-based methods.

2.1 Fourier-based methods

The Fourier-based methods are based on the idea that the stripe artifacts are specific structures that introduce specific patterns in the Fourier domain by introducing energy at a specific level of frequencies. Indeed, if we look at the Figure 2f, we can see that the stripe artifacts are represented by a line of high energy at a given position in the Fourier domain. So removing this line of high energy can help to remove the stripe artifacts from the sinogram and improve the quality of the reconstructed image, as shown in Figure 3b.

The challenge of the Fourier-based methods is to accurately identify the line of high energy corresponding to the stripe artifacts and to remove it without affecting the other components of the sinogram, which can lead to a loss of information and a degradation of the quality of the reconstructed image.

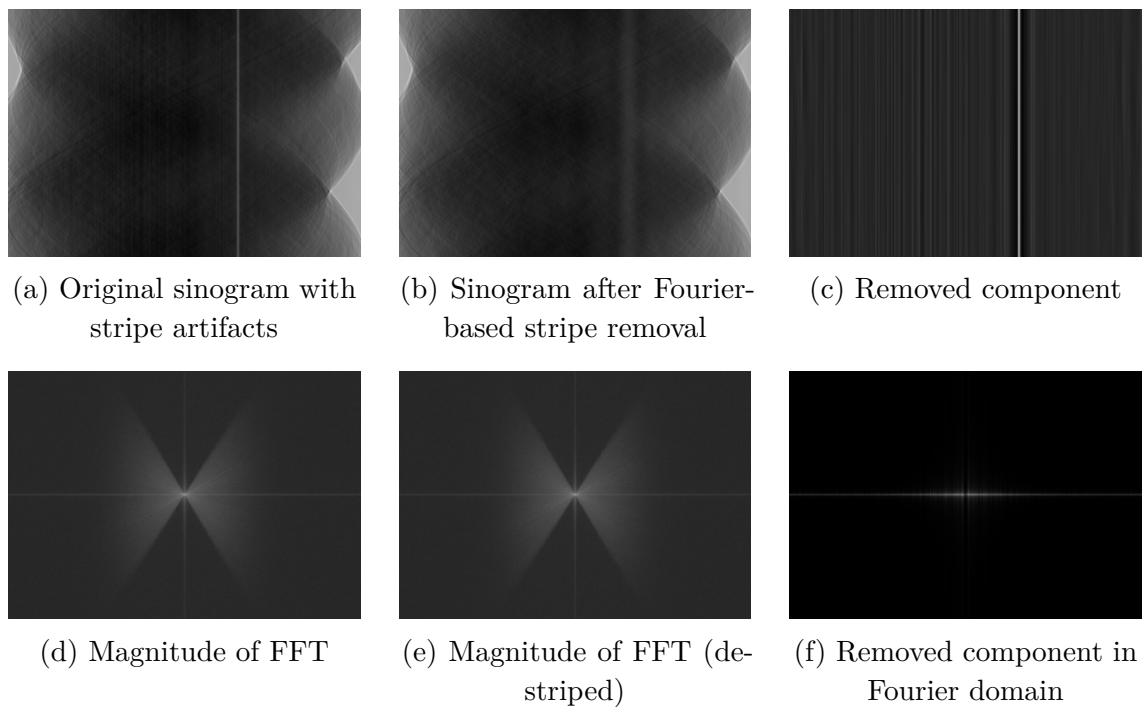


Figure 2: Fourier-based stripe removal: visualizing the process in spatial and Fourier domains

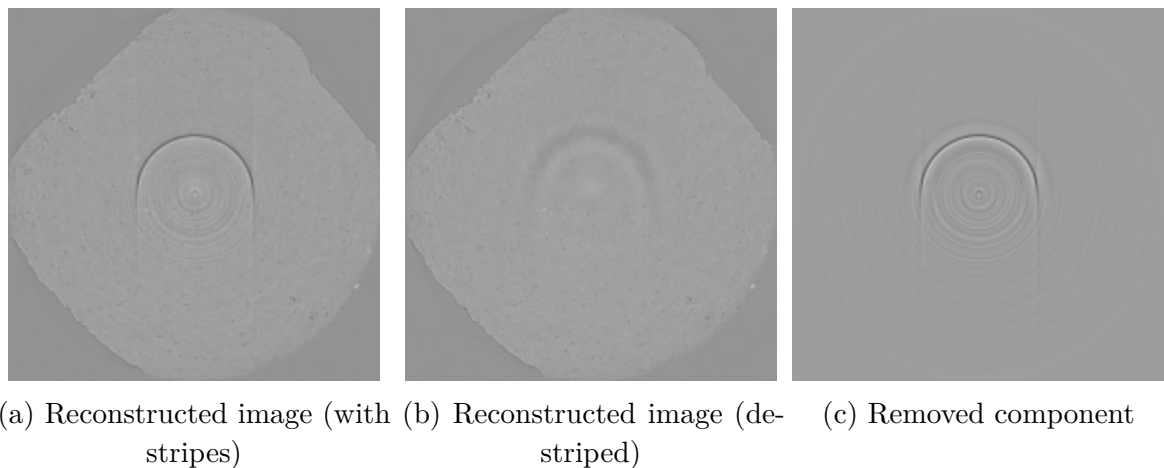


Figure 3: Effect of Fourier-based stripe removal on the reconstructed image

2.2 Wavelet-based methods

The wavelet-based methods are based on the idea that the stripe artifacts can be removed by applying a wavelet decomposition to the sinogram. Then, some of the wavelet coefficients capturing the stripe artifacts can be identified, like in Figure 4c. Applying the Fourier transform to these coefficients allows to identify the line of high energy corresponding to the stripe artifacts, like in Figure 4g. Removing this line of high energy and applying the inverse Fourier transform allows to remove the stripe artifacts from the wavelet coefficients. Then, applying the inverse wavelet transform allows to reconstruct the sinogram without stripe artifacts, as shown in Figure 4b.

This method, compared to the Fourier-based methods, allows to easily identify the high energy line corresponding to the stripe artifacts in the Fourier domain by looking at the wavelet coefficients.

The parameters of the wavelet decomposition, such as the type of wavelet, the level of decomposition and the strength of the removal, can affect the performance of the method and need to be carefully chosen to achieve good results without introducing artifacts or losing information in the reconstructed image.

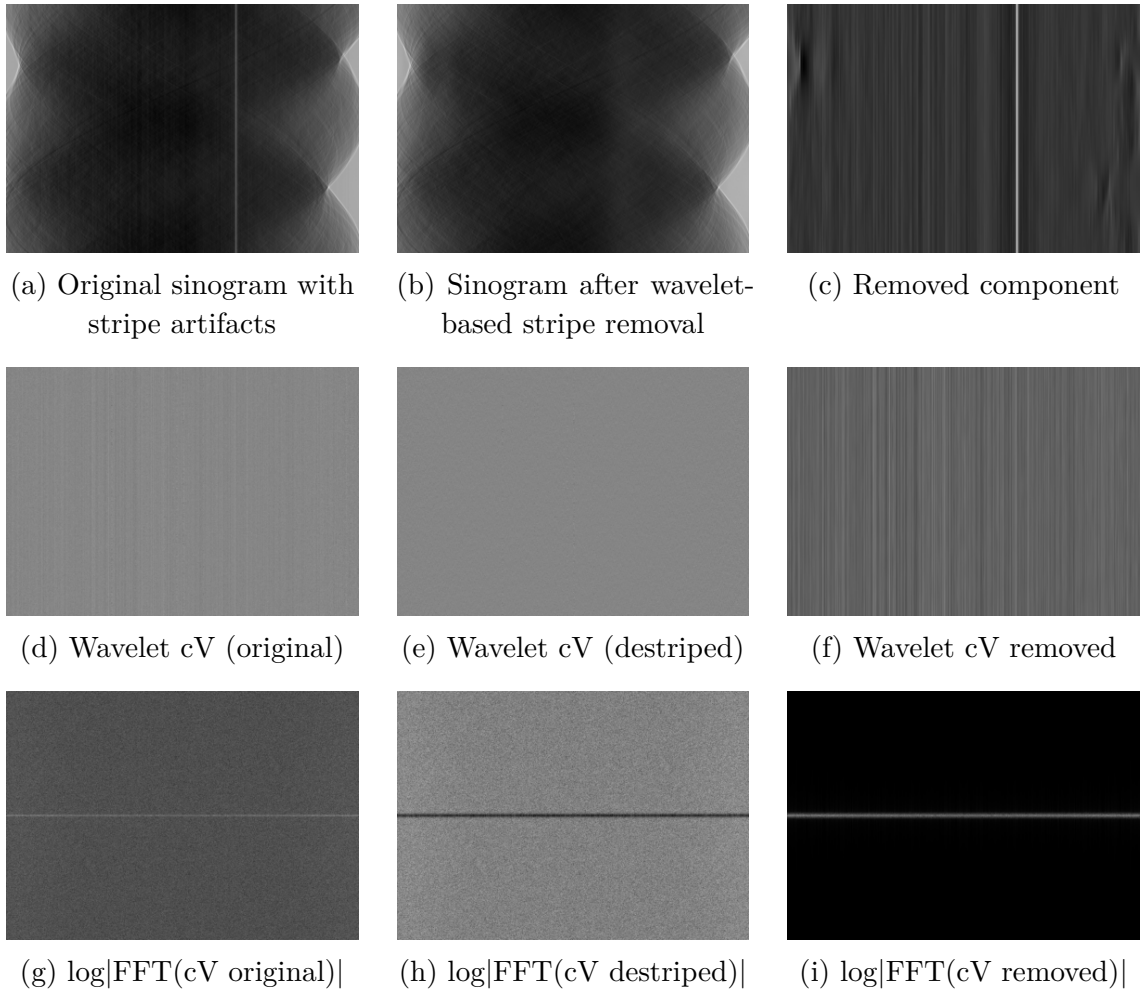
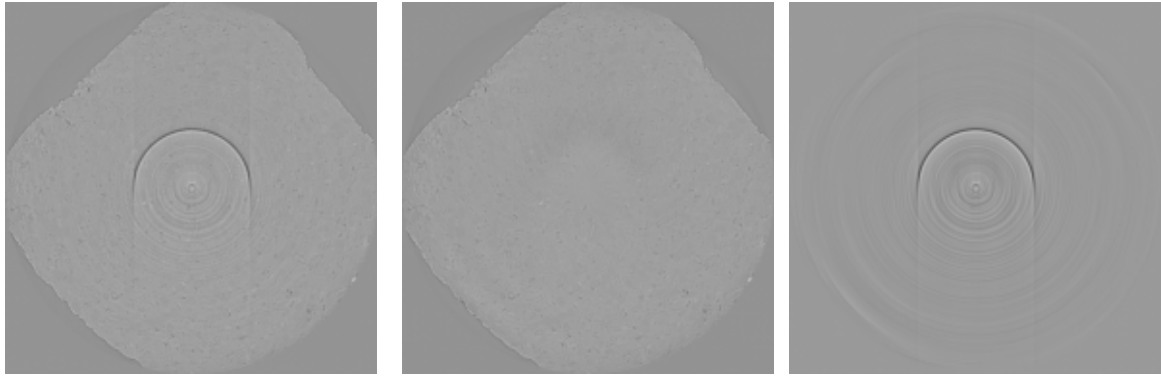


Figure 4: Wavelet-based stripe removal: visualizing the process in spatial, wavelet, and Fourier domains



(a) Reconstructed image (with stripes) (b) Reconstructed image (de-striped) (c) Removed component

Figure 5: Effect of wavelet-based stripe removal on the reconstructed image

2.3 Normalization-based methods

The normalization-based methods are based on the assumption that each column of the sinogram corresponds to a specific detector. If the detector is defective, the corresponding column will have a different mean or median value compared to the other columns. The normalization then consists to retrieve the good calibration of the detector by finding the baseline tendency of columns of the sinogram by analyzing their statistical properties, such as the mean or median. To do that, a Gaussian filter can be applied to the sinogram to smooth the columns and retrieve the baseline tendency by deleting the high frequency components corresponding to the noise that distort the estimation of the baseline tendency. The strength of the Gaussian filter impacts the performance of the method, as a too strong filter can lead to a loss of information and a degradation of the quality of the reconstructed image, while a too weak filter can lead to a poor estimation of the baseline tendency and thus a poor performance of the method.

So the normalization-based methods can help to reduce the intensity of the stripe artifacts by correcting the calibration of the detector and thus reducing the intensity of the stripe artifacts in the sinogram, which can improve the quality of the reconstructed image, as shown in Figure 6e.

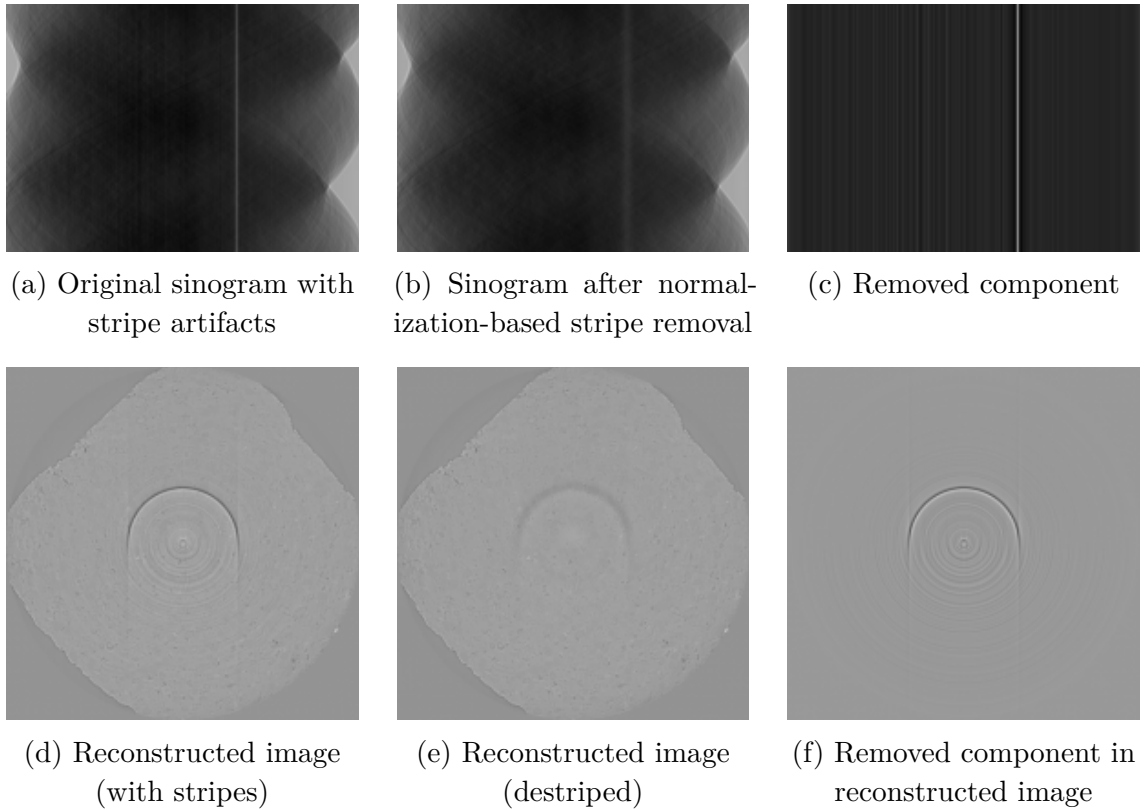


Figure 6: Normalization-based stripe removal

2.4 Regularization-based methods

The regularization-based methods are based on the idea that the stripe artifacts can be removed by adding a regularization term to the reconstruction problem.

Indeed, we can consider that the acquisition of the sinogram is a linear process that can be represented by a linear operator A , such that the sinogram y can be expressed as $y = A x + n$, where x is the original image and n is the noise. The reconstruction of the image from the sinogram can then be formulated as an optimization problem, where we want to find the image x that minimizes the difference between the sinogram y and the projection of the image $A x$, while also adding a regularization term $R(x)$ that imposes certain constraints on the solution. Indeed, the regularization term indicates which kind of solution we want to obtain, such as a smooth solution or a sparse solution.

So the main challenge of the regularization-based methods is to choose the right regularization term that can effectively remove the stripe artifacts while preserving the important features of the image. The method works quite well as shown in Figure 7.

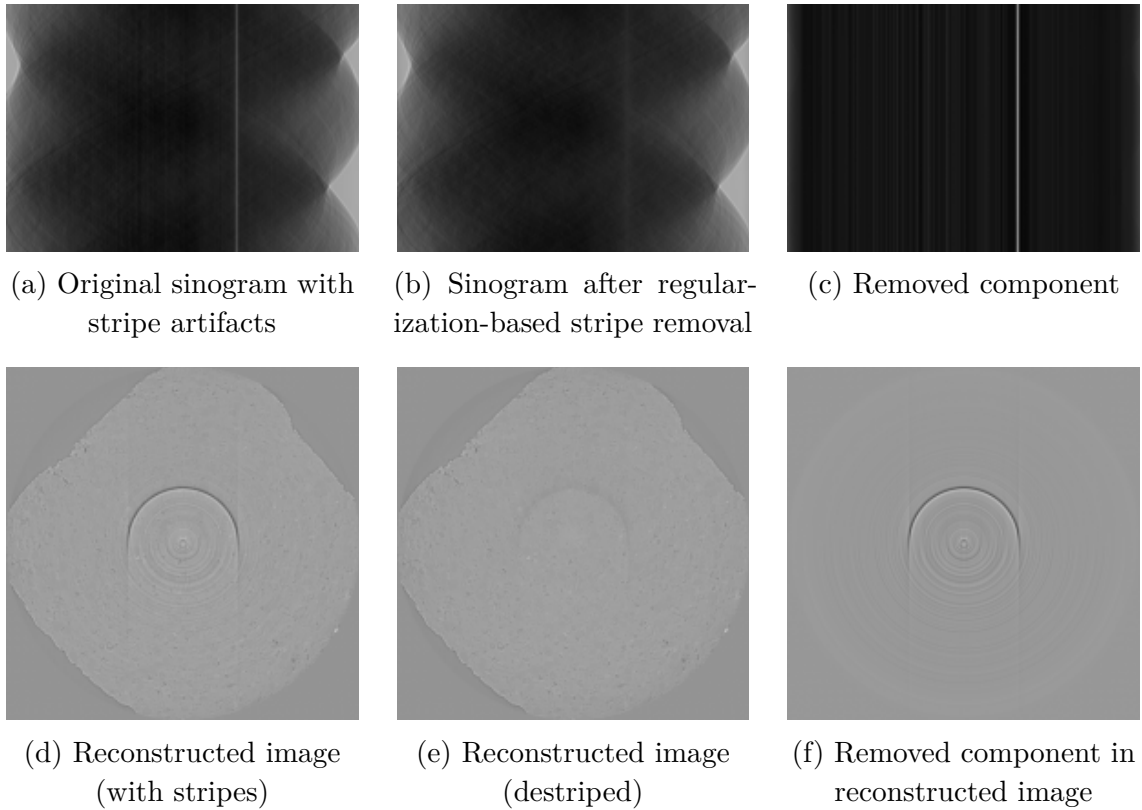
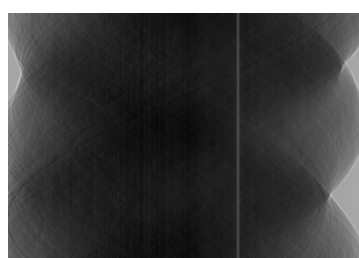


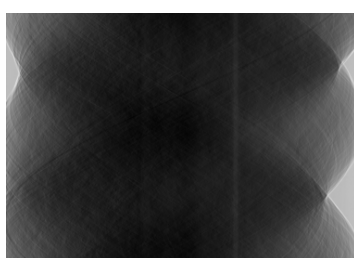
Figure 7: Regularization-based stripe removal

2.5 Sorting-based methods

The sorting-based methods are based on the idea that the stripe artifacts can be removed by sorting the columns of the sinogram based on their intensity values. The main idea is that the columns corresponding to the defective detectors will have much different intensity values compared to the other columns. So the sorting allows to identify these columns by sorting the intensity values inside each column and looking at the distribution of the sorted values across the columns. Then, the columns corresponding to the defective detectors can be identified by looking at the distribution of the sorted values across the columns. The distribution of defective columns can be equalized by applying some correction such as enforcing the same distribution of intensity values across the columns. To do that, a filter like a median filter can be applied to the sorted sinogram (shown in Figure 8d) to smooth the distribution of intensity values across the columns and thus reduce the intensity of the stripe artifacts in the sinogram, which can improve the quality of the reconstructed image, as shown in Figure 8h. Then, the inverse sorting can be applied to retrieve the destriped sinogram in the original order of columns.



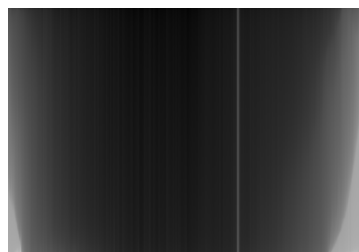
(a) Original sinogram with stripe artifacts



(b) Sinogram after sorting-based stripe removal



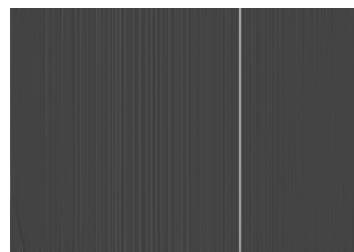
(c) Removed component



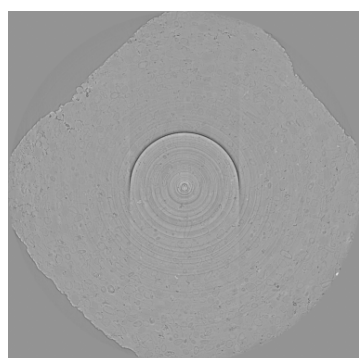
(d) Sorted sinogram



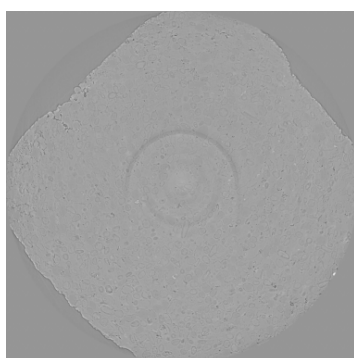
(e) Sorted destriped sinogram



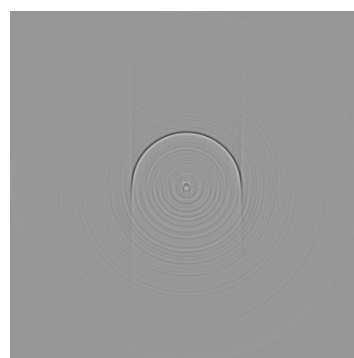
(f) Removed component in sorted sinogram



(g) Reconstructed image (with stripes)



(h) Reconstructed image (destriped)



(i) Removed component in reconstructed image

Figure 8: Sorting-based stripe removal



3 Conclusion

In this report, we have explored different methods to remove stripe artifacts from sinograms and reconstruct the image.

The Fourier-based methods are based on the idea that the stripe artifacts are specific structures that introduce specific patterns in the Fourier domain by introducing energy at a specific level of frequencies.

The wavelet-based methods are based on the idea that the stripe artifacts can be removed by applying a wavelet decomposition to the sinogram and removing the high energy line corresponding to the stripe artifacts in the Fourier domain of the wavelet coefficients.

The normalization-based methods are based on the assumption that each column of the sinogram corresponds to a specific detector and that the defective detectors will have different intensity values compared to the other columns, so the normalization consists to retrieve the good calibration of the detector by finding the baseline tendency of columns of the sinogram by analyzing their statistical properties, such as the mean or median.

The regularization-based methods are based on the idea that the stripe artifacts can be removed by adding a regularization term to the reconstruction problem.

The sorting-based methods are based on the idea that the stripe artifacts can be removed by sorting the columns of the sinogram based on their intensity values and applying some correction to equalize the distribution of intensity values across the columns.

The choice of the method to use depends on the type of stripe artifacts and requirements of the application, such as the quantity of information to preserve or some other criteria. The removal of stripe artifacts is very dependent on the parameters involved in the methods, such as the location of the high energy lines in the Fourier domain, the strength of the Gaussian filter for the normalization-based methods, the type of wavelet and level of decomposition for the wavelet-based methods, the regularization term for the regularization-based methods, and the size of the filter for the sorting-based methods.¹

¹ Usage of AI: ChatGPT to understand how to work with algotom and sinograms, and to summarize papers explaining the different methods for stripe removal, VSCode autocompletion for Python and report writing.